Sumit Goel Email: sumitgoel58@gmail.com | Website: goelsumit.com

Research Statement

I am a microeconomic theorist working on economic design. I use game theory to study how institutions shape individual behavior and social outcomes. My research aims to develop implementable solutions to practical design problems and contributes to three broad areas: allocation problems, contest theory, and mechanism design without money.

Allocation problems. Allocating resources fairly and efficiently is a classical problem in economics and computer science. A major challenge is that agents' preferences over allocations are often their private information, so allocation rules must also be robust to strategic behavior. Unfortunately, these goals are often incompatible, especially when monetary transfers cannot be used to compensate agents. A natural direction then is to understand the extent to which these goals can be met, which is central to my research agenda.

An important instance of this problem is object reallocation, wherein each agent is endowed with a single object and objects can be reallocated among agents, as in kidney exchange. The Top Trading Cycles (TTC) rule is fundamental to this problem: it satisfies a host of desirable properties, and it is the only rule satisfying the relatively modest requirements of individual rationality, pair efficiency, and strategyproofness on the unrestricted domain. Even so, in [9], we illustrate how TTC may select unappealing allocations, and ask whether one can move beyond TTC on restricted domains. We show that non-TTC rules satisfying these properties exist precisely on domains failing our *top-two condition*, and construct a rule that overcomes the aforementioned shortcoming. When each agent has multiple objects, a deeper problem arises: even stable allocations that are immune to coalitional blocks may fail to exist. In [2], we identify conditions under which such allocations exist and propose a generalized TTC algorithm that finds an allocation in the stable set.

Other work studies the allocation of objects over multiple periods, such as dorm rooms to students each year. In [3], we introduce *swap bounded envy*, a relaxation of envy-freeness under which any envy can be eliminated by a single swap of objects, and propose a TTC+Serial Dictatorship algorithm that produces efficient allocations satisfying it. When monetary transfers are possible, auctions are a widely used institution for allocating resources. In [10], we establish an ordering of k-price auctions by their worst-case allocative efficiency.

In future work, I plan to study the design of allocation rules in the presence of externalities and behavioral biases. An ongoing project, set in a matching-with-contracts environment, asks whether a decentralized implementation of the cumulative offer process preserves the stability and incentive guarantees of its centralized counterpart when agents are behavioral.

Contest theory. Contests are a widely used institution for pursuing a variety of objectives: platforms run crowdsourcing events to solicit innovative solutions, firms run sales contests to boost revenue, and schools use exams to screen students. Contests also arise naturally through market forces, as in R&D races. Across these examples, the participating agents exert effort so as to outperform others and win better rewards. My research studies how the underlying structure of a contest shapes these incentives to exert effort and, given the designer's objective, which structure is optimal.

The most basic structural element of a contest is its prize structure: how a fixed budget is divided among prizes for different ranks. A natural question is whether making prizes more unequal, and thus the contest more competitive, encourages effort. In [1], we develop a novel approach to analyzing contests when agents have one of finitely many types, providing a unified perspective on the contrasting results in complete-information and continuum-type environments. We show that shifting value to the top prize generally encourages aggregate effort, because it strongly motivates the most productive agents. Shifting value to an intermediate prize, however, may discourage effort, because it weakens incentives to compete for the top prize. As a consequence, the winner-takes-all contest is robustly optimal: under linear, concave, and even moderately convex effort costs, it maximizes the total effort of the top q agents for any q. Prizes need not be monetary: in classrooms, grades are valuable for the information they convey about ability. In [4], I show that making grades more informative is akin to making prizes more unequal. Within the scope of contests induced by grading schemes, more informative grading encourages effort when agents of moderate ability are likely and discourages it when they are unlikely.

In some settings, such as programming competitions, agents cannot be ranked by effort or output, because the designer observes only a coarse measure of performance. For such environments, in [8], we introduce and demonstrate the desirability of *Luce contracts*, which assign weights to agents and distribute the budget among successful agents in proportion to these weights. Other work focuses on two-player contests, where the choice of prize structure is trivial, and explores the design of other structural elements: in [5], we study tie-breaking rules in Tullock contests with ties, and in [11], we examine feedback policies in dynamic all-pay auctions.

These papers, and the contest literature more broadly, point to a rich theory of how uncertainty about agents' abilities, the shape of their effort cost functions, and the noise in their observed outputs jointly determine how contest structure shapes incentives. In future work, I aim to better understand how these forces interact.

Mechanism design without money. For a remarkably broad family of preference aggregation problems, the Vickrey–Clarke–Groves (VCG) mechanism offers a simple recipe for implementing efficient outcomes while ensuring strategyproofness: it uses monetary transfers to align agents' incentives with those of society. Despite its generality, VCG has

shortcomings. It may be computationally intractable, as in combinatorial auctions, and monetary transfers may be impractical, as in public decision making, rendering VCG infeasible. Focusing on the second concern, my research explores how close we can get to efficient outcomes while maintaining strategyproofness without the flexibility of transfers.

A canonical setting for studying this question is facility location. On a line with Euclidean preferences, the median rule is known to be both strategyproof and efficient, so the absence of transfers is costless. In [6], we study facility location in the plane, where this is no longer the case: we show that the coordinate-wise median mechanism is optimal among strategyproof mechanisms for a broad class of social cost functions, and quantify the resulting welfare loss.

In [7], we study a different setting: a principal–agent project selection problem with asymmetric information, where the principal is powerless in the absence of transfers. Nevertheless, we show that if the principal can induce or identify partial verifiability, so that certain misreports are infeasible, there are simple mechanisms that perform well. For instance, suppose the agent cannot oversell any project, perhaps because it must provide evidence in support of its claims, so that the principal need only guard against underselling. Then a mechanism that selects the agent's most preferred project among those meeting a payoff threshold for the principal is optimal.

While my existing work insists on exact strategyproofness, an ongoing project explores whether simply choosing the efficient outcome is Probably Approximately Strategyproof (PAS)—a notion of approximate incentive compatibility inspired by the Probably Approximately Correct (PAC) framework in machine learning.

References

[1] Andrzej Baranski and Sumit Goel. The effect of competition in contests: A unifying approach. Working paper, 2024.

[2] Federico Echenique, Sumit Goel, and SangMok Lee. Stable allocations in discrete exchange economies. *Journal of Economic Theory*, 222:105921, 2024.

[3] Federico Echenique, Sumit Goel, and SangMok Lee. Swap bounded envy. *arXiv preprint arXiv:2508.09290*, 2025.

[4] Sumit Goel. Optimal grading contests. *Games and Economic Behavior*, 152:133–149, 2025.

[5] Sumit Goel and Amit Goyal. Optimal tie-breaking rules. *Journal of Mathematical Economics*, 108:102872, 2023.

[6] Sumit Goel and Wade Hann-Caruthers. Optimality of the coordinate-wise median mechanism for strategyproof facility location in two dimensions. *Social Choice and Welfare*, 61(1):11–34, 2023.

[7] Sumit Goel and Wade Hann-Caruthers. Project selection with partially verifiable information. *Mathematical Social Sciences*, 132:105–113, 2024.

[8] Sumit Goel and Wade Hann-Caruthers. Multi-agent contract design with a budget. *Games and Economic Behavior*, page 104041, 2026.

[9] Sumit Goel and Yuki Tamura. Ttc domains. Working paper, 2025.

[10] Sumit Goel and Jeffrey Zeidel. An efficiency ordering of k-price auctions under complete information. *Economics Letters*, 255:112565, 2025.

[11] Sumit Goel, Jeffrey Zeidel, and Yiqing Yan. Feedback in dynamic contests: Theory and experiment. *arXiv preprint arXiv:2510.23178*, 2025.